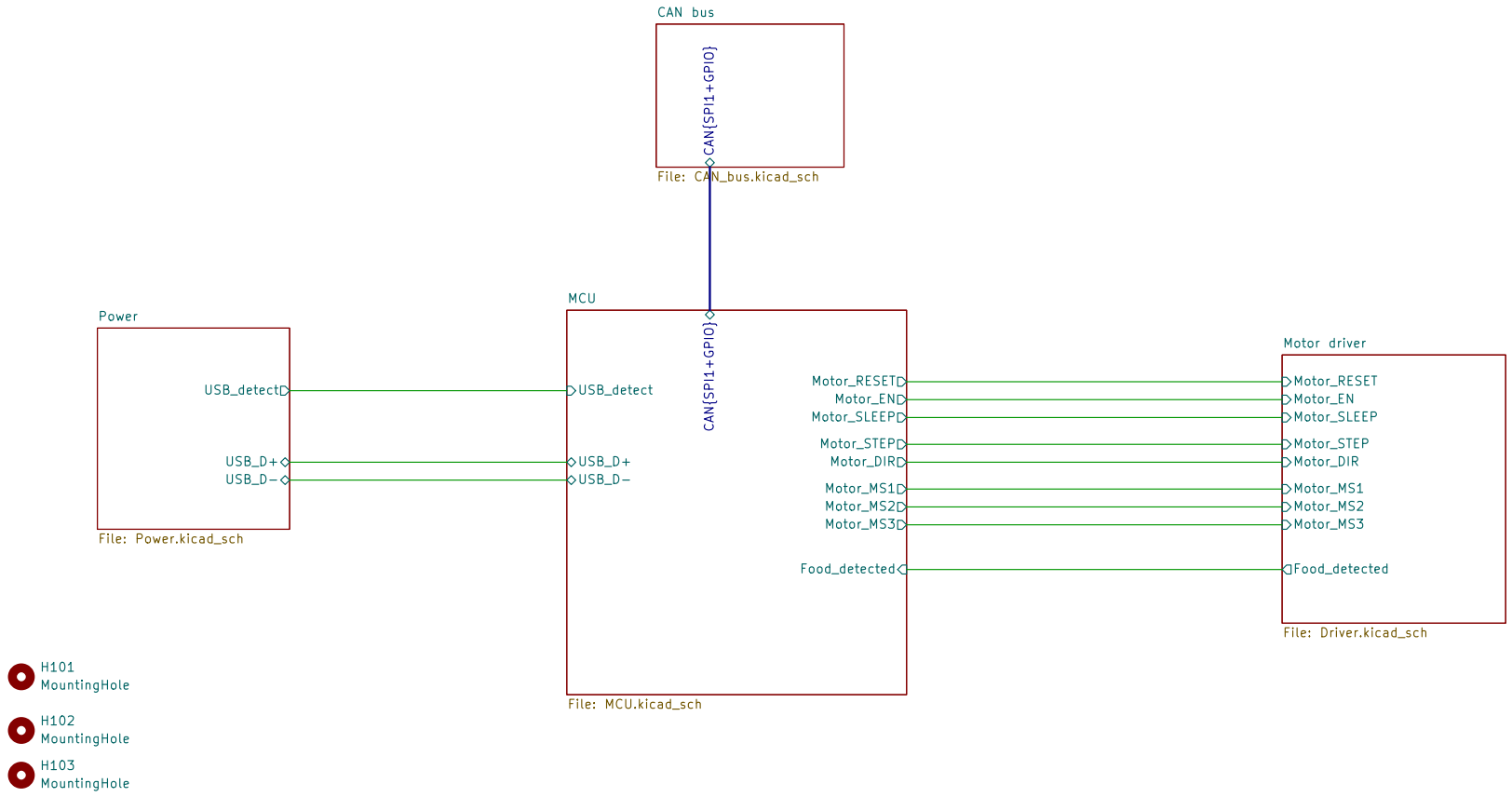
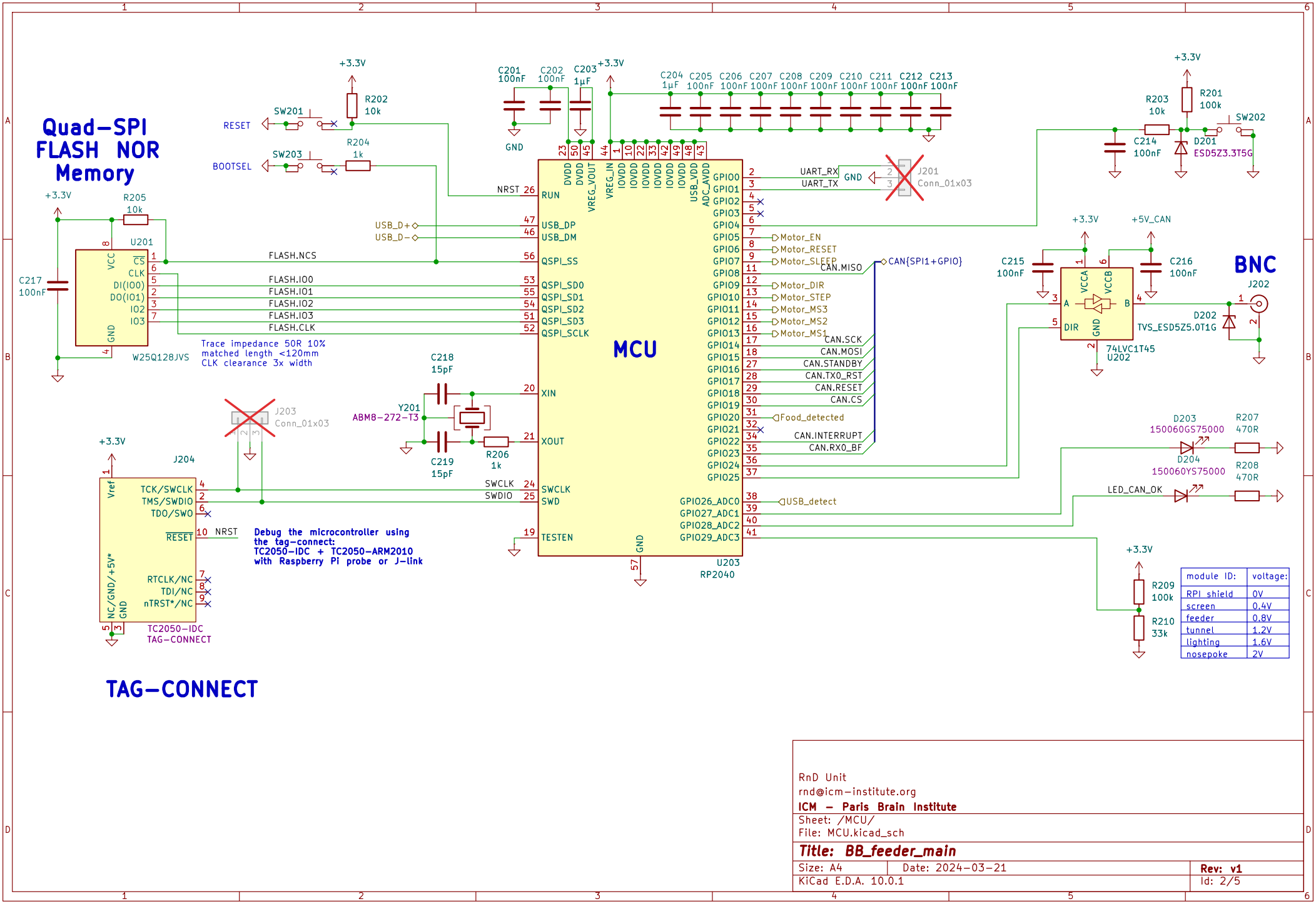


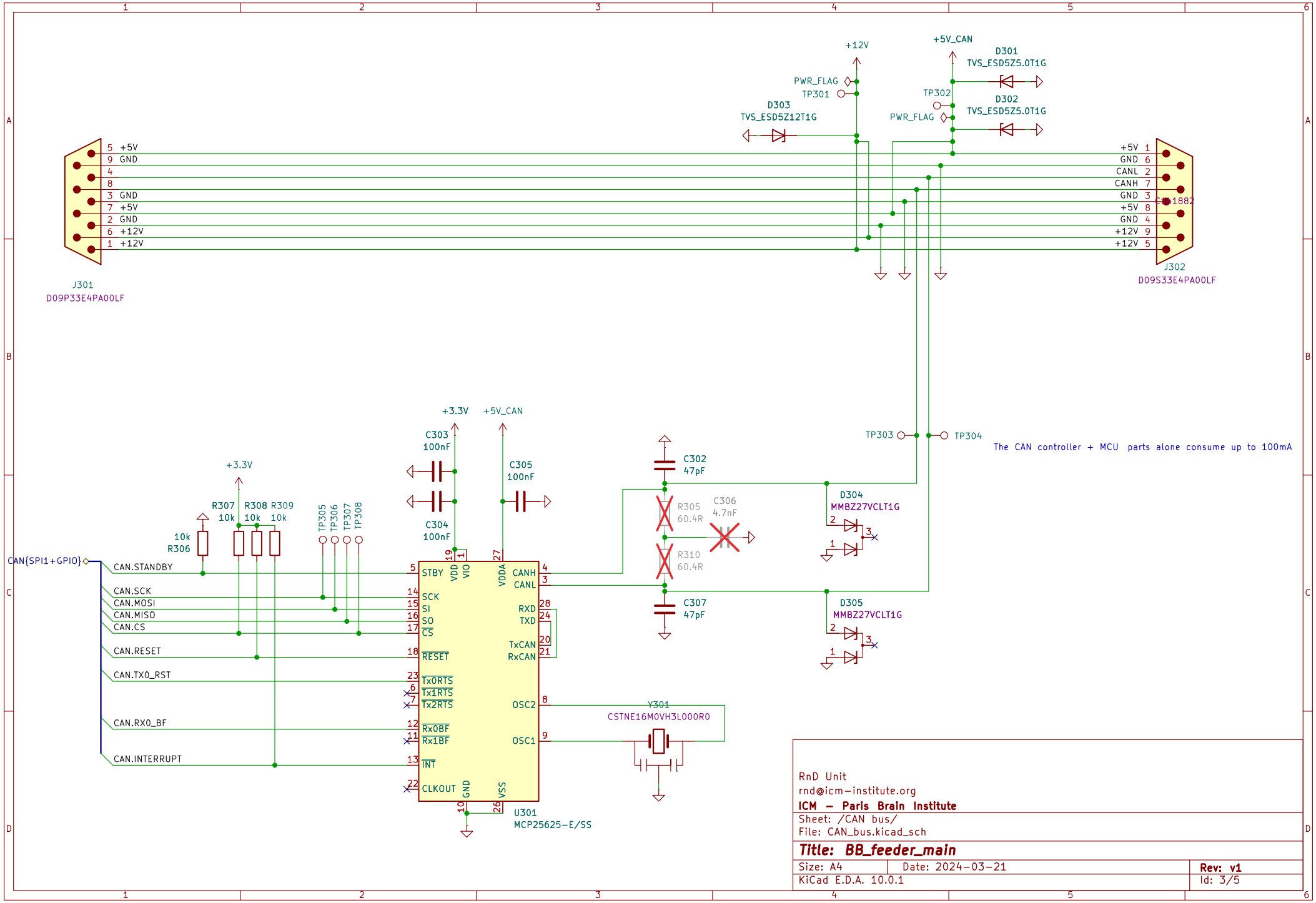
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute

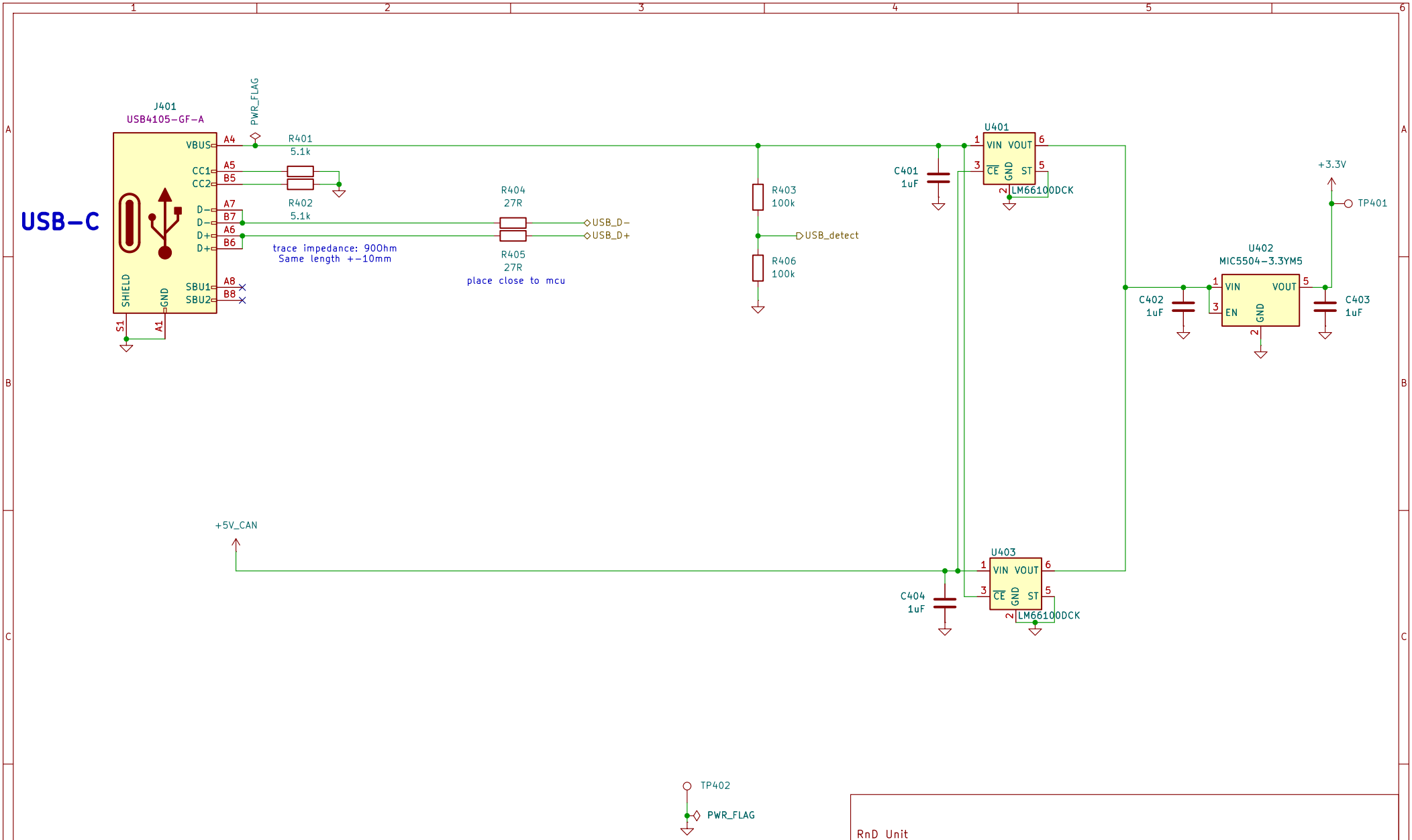
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File: BB_feeder_main_V1.kicad_sch

Title: BB_feeder_main

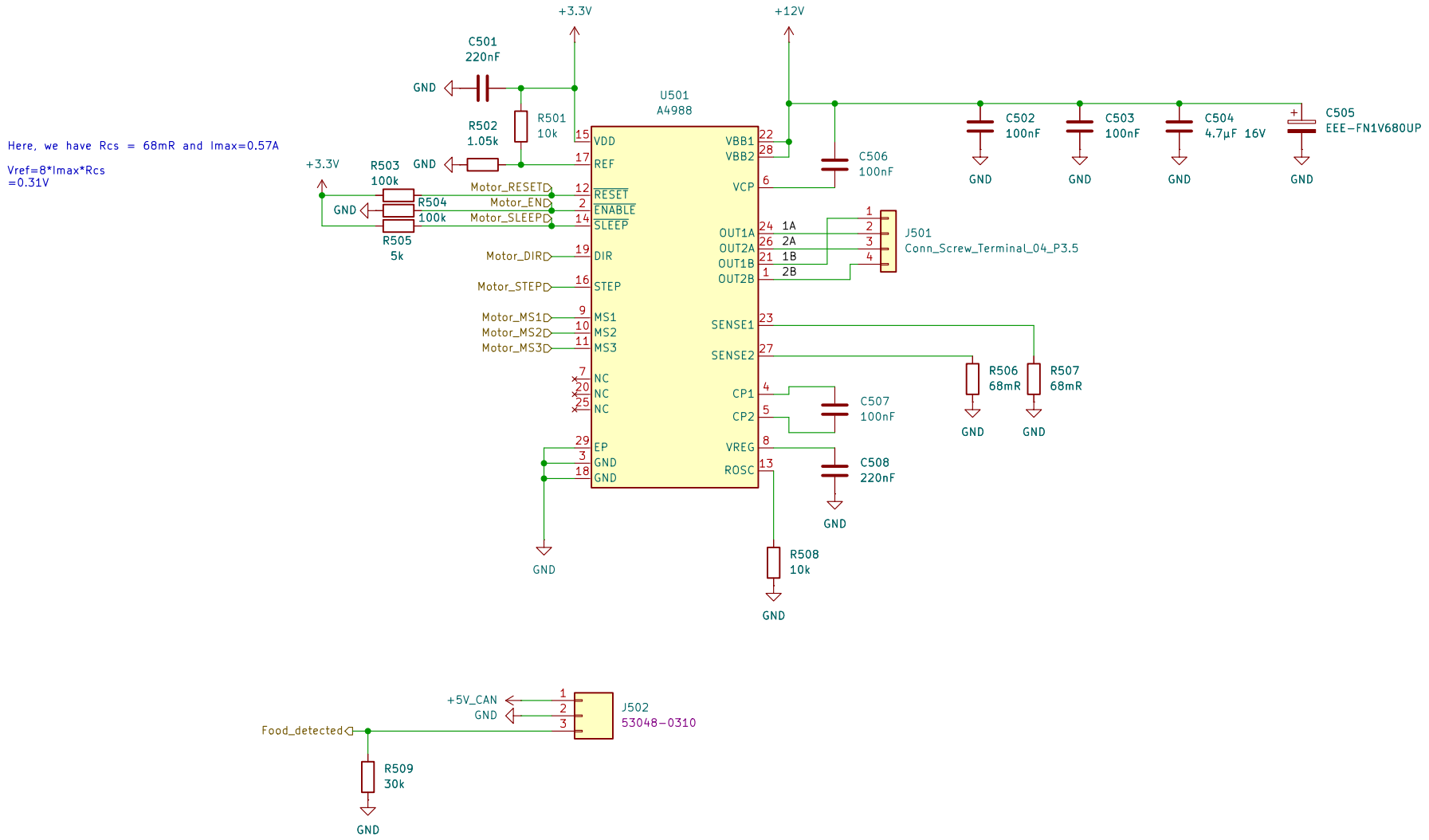
Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 1/5


Quad-SPI FLASH NOR Memory
W25Q128JVS
Trace impedance 50R 10%
matched length <120mm
CLK clearance 3x width
MCU
RP2040
Debug the microcontroller using the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
TC2050-IDC
TAG-CONNECT
BNC
74LVC1T45
TVS_ESD5Z5.0T1G
ESD5Z3.3T5G
150060GS75000
150060YS75000
module ID: | voltage:
RPI shield | 0V
screen | 0.4V
feeder | 0.8V
tunnel | 1.2V
lighting | 1.6V
nosepoke | 2V
RnD Unit
rnd@icm-institute.org
ICM - Paris Brain Institute
Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_feeder_main
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 2/5


+12V
+5V_CAN
D301
TVS_ESD5Z5.0T1G
PWR_FLAG
TP301
TP302
D303
TVS_ESD5Z12T1G
D302
TVS_ESD5Z5.0T1G
PWR_FLAG
5 +5V
9 GND
4
8
3 GND
7 +5V
2 GND
6 +12V
1 +12V
+5V 1
GND 6
CANL 2
CANH 7
GND 3
+5V 8
GND 4
+12V 9
+12V 5
J301
D09P33E4PA00LF
J302
D09S33E4PA00LF
TP303
TP304
The CAN controller + MCU parts alone consume up to 100mA
+3.3V
+5V_CAN
C303
100nF
C305
100nF
C304
100nF
C302
47pF
R305
60.4R
C306
4.7nF
R310
60.4R
C307
47pF
D304
MMBZ27VCLT1G
D305
MMBZ27VCLT1G
+3.3V
R307
10k
R308
10k
R309
10k
10k
R306
TP305
TP306
TP307
TP308
CAN{SPI1+GPIO}
CAN.STANDBY
CAN.SCK
CAN.MOSI
CAN.MISO
CAN.CS
CAN.RESET
CAN.TX0_RST
CAN.RX0_BF
CAN.INTERRUPT
STBY
SCK
SI
SO
CS
RESET
TX0RTS
TX1RTS
TX2RTS
RX0BF
RX1BF
INT
CLKOUT
VDD
VIO
GND
VDDA
CANH
CANL
RXD
TXD
TxCAN
RxCAN
OSC2
OSC1
VSS
Y301
CSTNE16M0VH3L000R0
U301
MCP25625-E/SS
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /CAN bus/
File: CAN_bus.kicad_sch
Title: BB_feeder_main
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 3/5



RnD Unit	
rnd@icm-institute.org	
ICM – Paris Brain Institute	
Sheet: /Power/	
File: Power.kicad_sch	
Title: BB_feeder_main	
Size: A4	Date: 2024-03-21
KiCad E.D.A. 10.0.1	Id: 4/5
	Rev: v1


Here, we have Rcs = 68mR and Imax=0.57A
Vref=8*Imax*Rcs
=0.31V
+3.3V
+12V
C501
220nF
GND
R501
10k
R502
1.05k
R503
100k
R504
100k
R505
5k
U501
A4988
VDD
REF
RESET
ENABLE
SLEEP
DIR
STEP
MS1
MS2
MS3
NC
NC
NC
EP
GND
GND
VBB1
VBB2
VCP
OUT1A
OUT2A
OUT1B
OUT2B
SENSE1
SENSE2
CP1
CP2
VREG
ROSC
Motor_RESET
Motor_EN
Motor_SLEEP
Motor_DIR
Motor_STEP
Motor_MS1
Motor_MS2
Motor_MS3
C502
100nF
C503
100nF
C504
4.7µF 16V
C505
EEE-FN1V680UP
C506
100nF
J501
Conn_Screw_Terminal_04_P3.5
C507
100nF
C508
220nF
R506
68mR
R507
68mR
R508
10k
+5V_CAN
GND
J502
53048-0310
Food_detected
R509
30k
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /Motor driver/
File: Driver.kicad_sch
Title: BB_feeder_main
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 5/5